

Patent Search

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Abstract:
DESIGN AND DEVELOPMENT OF AN INTELLIGENT WEARABLE GLOVE SYSTEM ABSTRACT Mastering precise hand gestures is vital across domains such as education, rehabilitation, industrial control, and immersive virtual environments. Traditional training methods often suffer from delayed feedback, limited monitoring, and subjective evaluation. To address these challenges, this project presents an intelligent, sensor-integrated wearable glove designed to analyze, validate, and guide users through complex sequential gestures in real time. The system offers a holistic approach to gesture-based learning by combining advanced sensing technology, feedback mechanisms, and adaptive control algorithms. CD D) re 0. CD The glove is embedded with an array of flex sensors along each finger and inertial measurement units (IMUs) on the wrist and back of the hand, enabling precise tracking of bending, orientation, and motion dynamics. At its computational core lies a Finite State Machine (FSM)-based microcontroller, which continuously interprets gesture data, compares it with predefined sequences, and provides instant feedback through vibratory motors and LED indicators. This ensures accurate motion learning while minimizing human error. The FSM algorithm allows the glove to function in both autonomous and guided modes, adapting sensitivity and response based on user proficiency. CM E o ir in CM Tf Tf in CM o cs in o> oo Through Bluetooth or Wi-Fi connectivity, the glove interfaces with mobile or PC applications that visualize gestures, record performance analytics, and support gamified progression for skill enhancement. Furthermore, the system can communicate with external devices such as robotic arms, AR/VR platforms, and IoT-enabled tools, expanding its application potential. By merging sensing, intelligent state-based control, and real-time interactivity, the proposed glove establishes a novel standard for adaptive human-machine collaboration and experiential learning.

Complete Specification

FIELD OF INVENTION [001] The invention described relates to the field of wearable electronics and human-computer interaction, with particular applications in gesture recognition and control systems. More specifically, it falls within the following areas: [002] Wearable Gesture Recognition Systems: This includes designing smart gloves capable of detecting and interpreting hand gestures using integrated sensors and embedded processing for realtime applications. [003] Sensor Technology: This category includes flex sensors, IMU (accelerometer and gyroscope) sensors, and other motion-detecting devices, all integrated into the glove for precise measurement of finger bending, hand orientation, and motion dynamics. [004] Finite State Machine-Based Sequence Detection: The application of FSM logic for recognizing predefined gesture sequences allows robust and structured detection of dynamic gestures, enabling accurate interpretation of complex multi-gesture commands. CD ��> re 0. CD CM E o u_ m CM Tf

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